Computer Networks

Fall 2024 Semester

Course Code: CS – 3001

Offered to BSCS22, BDS21

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Lecture 3

August 27, 2024 (Tuesday)

Today's Agenda

- Networking Basics
  - Network Types
  - Network Topologies
  - Client-Server Model vs P2P Model
  - What is a protocol?
  - OSI Reference Model and TCP/IP Model
  - IP Addressing
  - Network Devices
  - Bandwidth and Latency
  - NAT

What is Networking?

- Networking is the practice of connecting computers and other devices to
  - share resources,
  - communicate, and
  - exchange data.

- Understanding the basic concepts of networking is crucial for designing, implementing, and maintaining networks efficiently and securely.

Network Overview

- What must a network provide?
  - connectivity
  - cost-effective sharing
  - functionality

- Performance
  - How are networks designed and built?
  - layering
  - protocols
  - standards

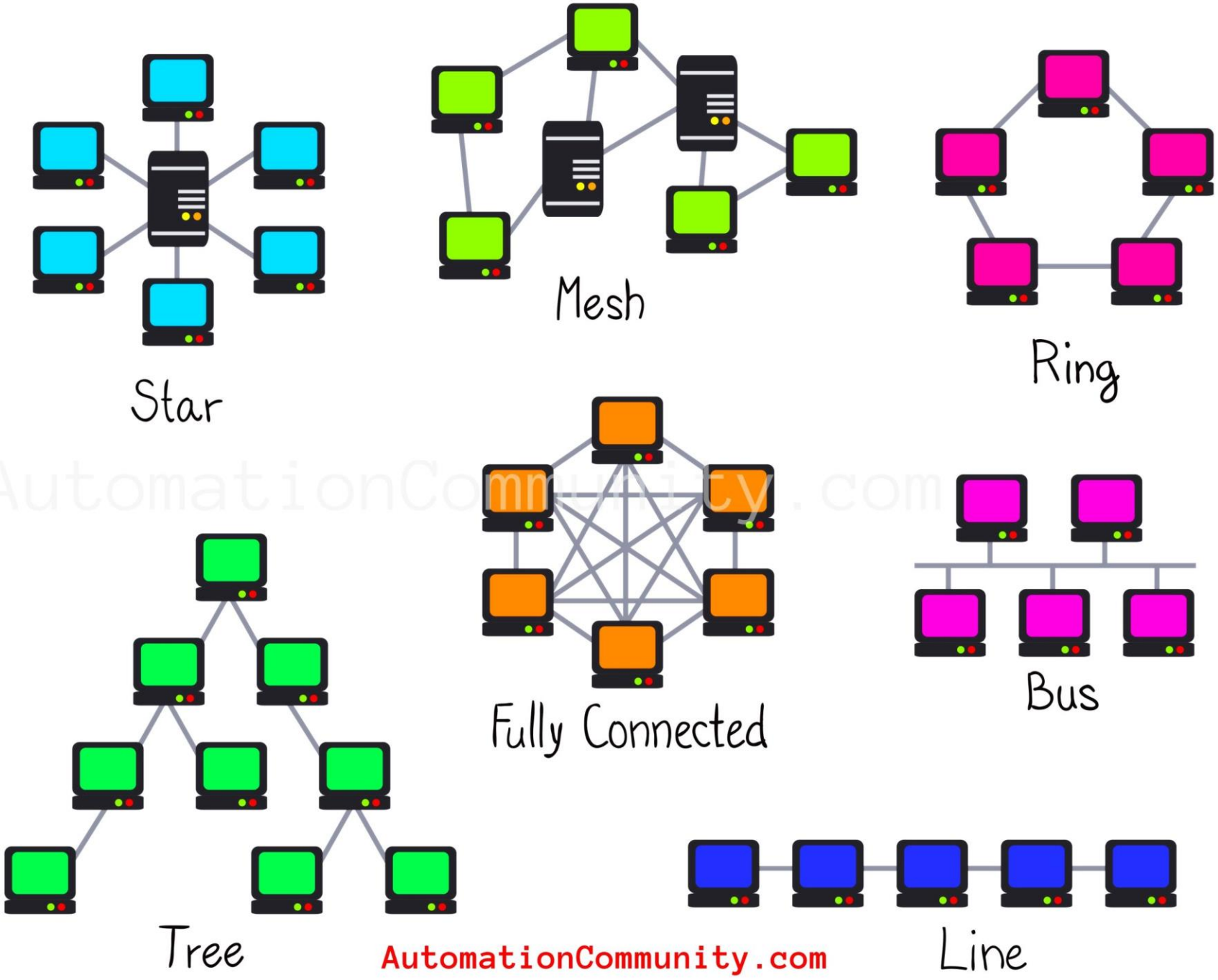
Network Types

- **Local Area Network (LAN):** A network that covers a small geographical area, like a home, office, or building.
- **Wide Area Network (WAN):** A network that covers a large geographical area, often composed of multiple interconnected LANs. The internet is the largest WAN.
- **Metropolitan Area Network (MAN):** A network that covers a city or a large campus.
- **Personal Area Network (PAN):** A network for personal devices, typically within a range of a few meters, like Bluetooth connections.

Network Topologies

- **Bus Topology:** All devices share a single communication line. Easy to implement but has performance and reliability drawbacks.
- **Star Topology:** All devices connect to a central hub or switch. If the central hub fails, the network is affected.
- **Ring Topology:** Devices are connected in a circular fashion, where each device is connected to two others. Data travels in one or both directions.
- **Mesh Topology:** Every device is connected to every other device, providing high redundancy and reliability but is complex and expensive.

Network Topologies



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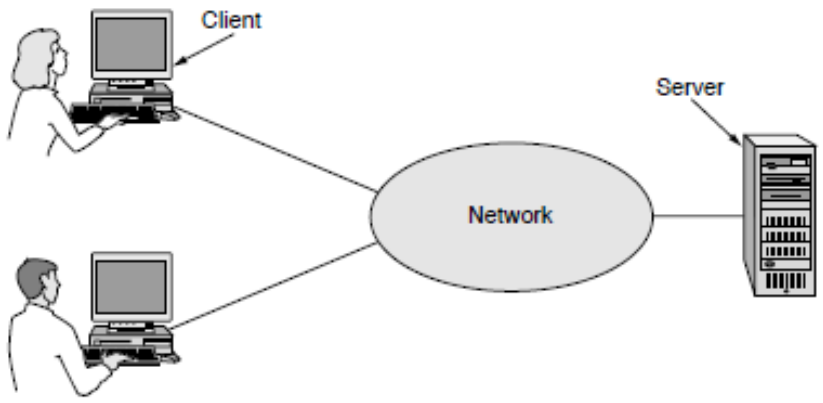
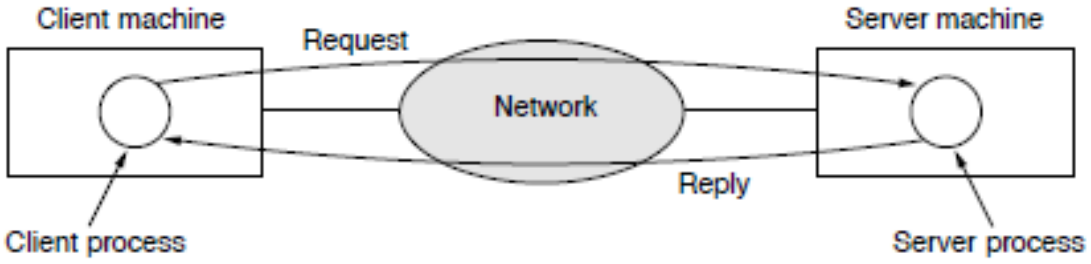
Client Server Model

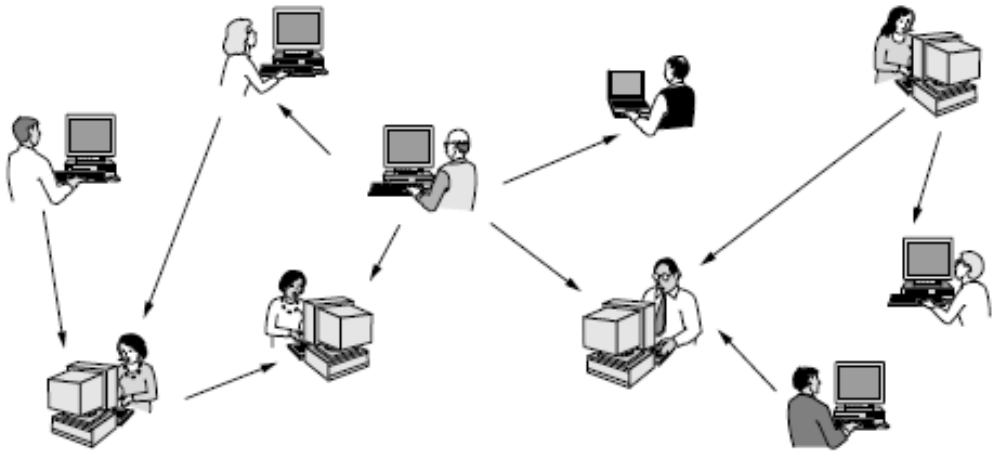
- The client-server model is a foundational concept in networking and computing.
- This model describes how devices or applications (clients) interact with a central server to share resources, process data, or perform tasks.
  - **Clients** are typically end-user devices (PCs, smartphones, or applications that access the server).
  - **server** is a powerful computer or software application that provides services, resources, or data to clients upon request.





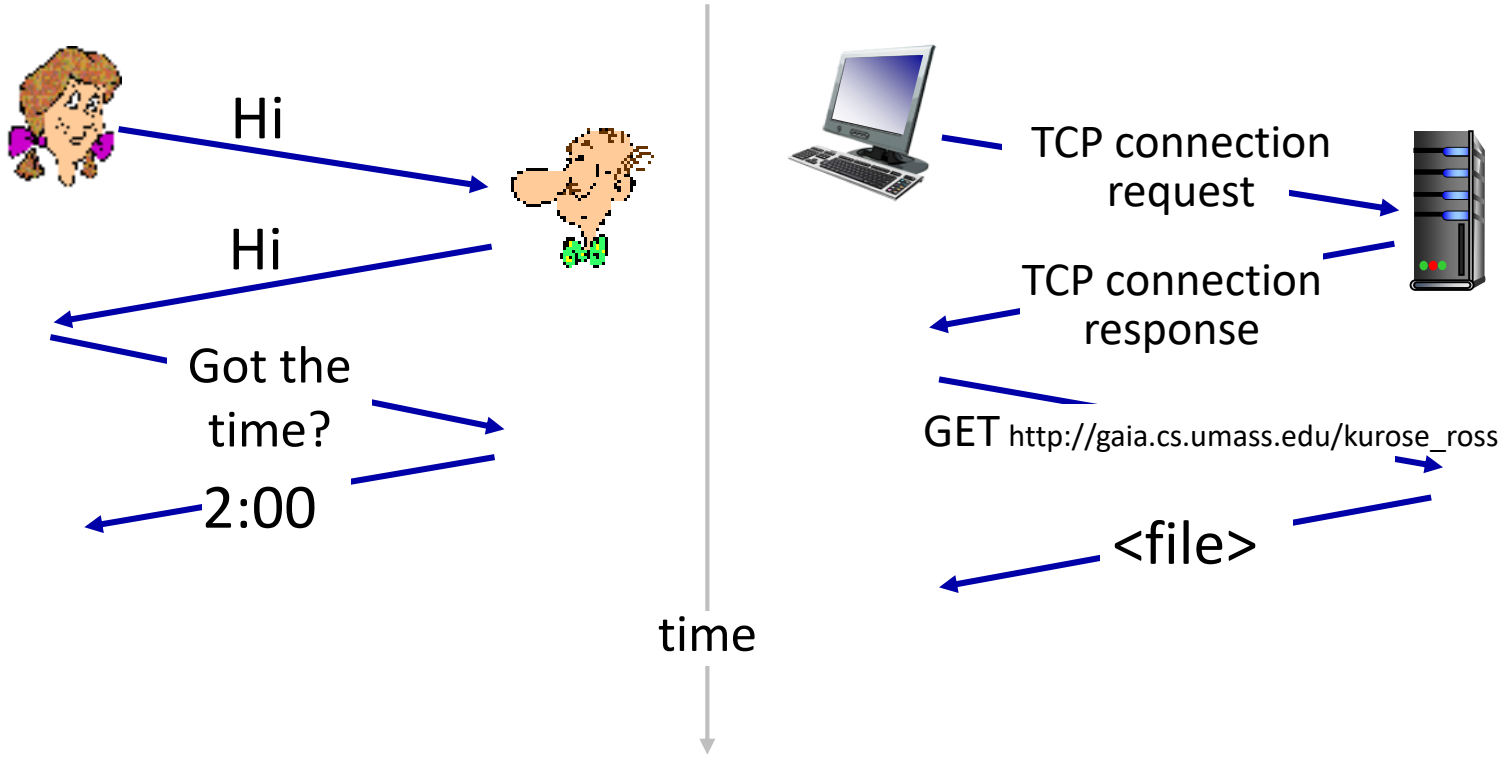
P2P (Peer-to-Peer) Model

- In this model, each device can act as both a client and a server, sharing resources directly with each other without a centralized server.
- Each user maintains a local DB of content, as well as a list of other members of the system.
- A new user can then go to any existing member to see what he has and get the names of other members to inspect for more content and more names.



What is a protocol?

A human protocol and a computer network protocol:



Q: other human protocols?

What is a protocol?

Human protocols:

- "what's the time?"
- "I have a question"
- introductions

Rules for:

… specific messages sent

… specific actions taken when message received, or other events

Network protocols:

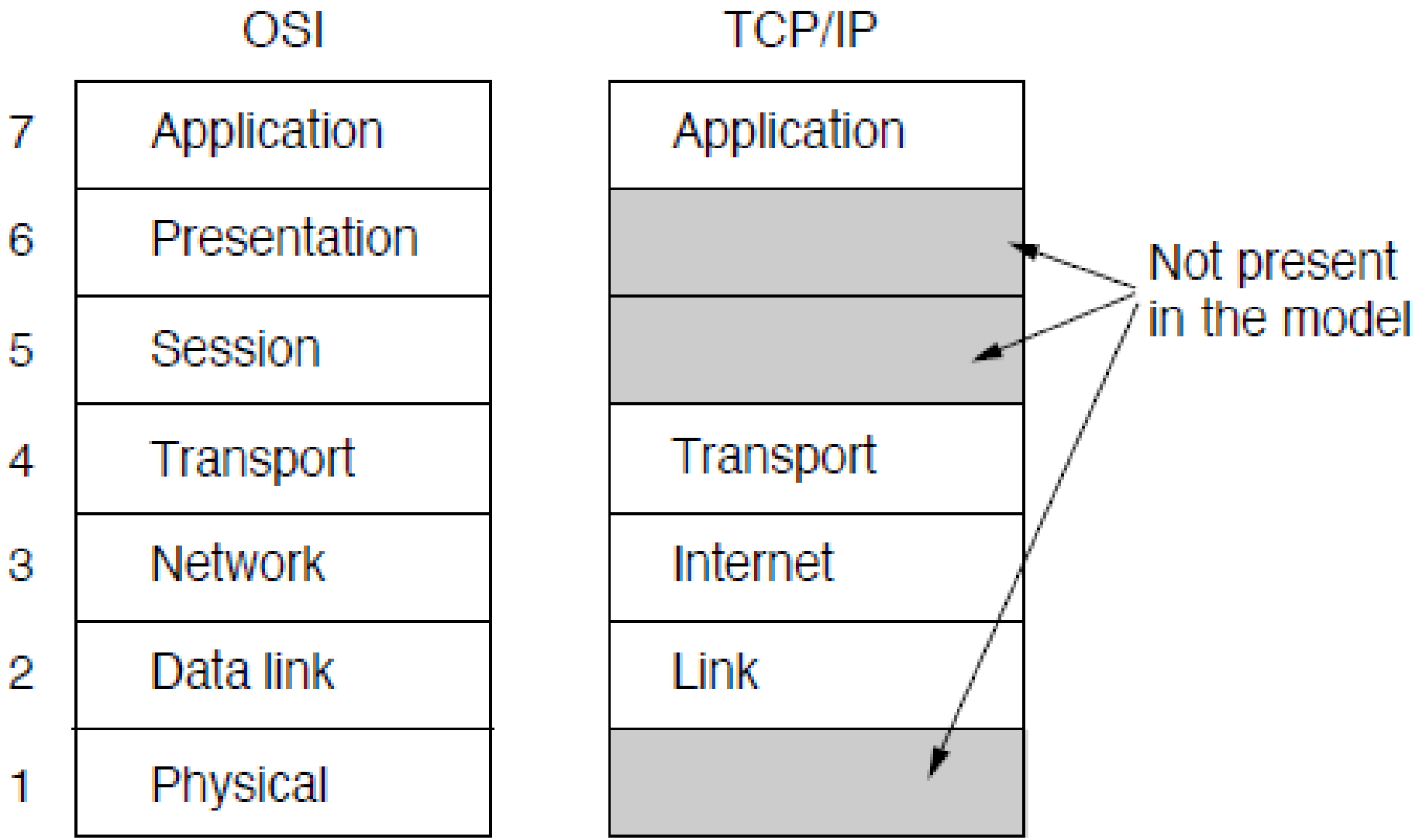
- computers (devices) rather than humans
- all communication activity in Internet governed by protocols

Protocols define the format, order of messages sent and received among network entities, and actions taken on message transmission, receipt

OSI Reference Model

- A conceptual framework used to understand how different networking protocols interact. It has seven layers:
  - **Physical Layer:** Deals with the physical connection between devices.
  - **Data Link Layer:** Handles error detection and correction from the Physical layer.
  - **Network Layer:** Manages device addressing and routes packets between networks.
  - **Transport Layer:** Ensures complete data transfer with error checking and flow control.
  - **Session Layer:** Manages sessions or connections between applications.
  - **Presentation Layer:** Ensures data is in a usable format and handles encryption/decryption.
  - **Application Layer:** The interface for end-user applications..

TCP/IP Reference Model



IP Addressing

- **IPv4:** A 32-bit address format, e.g., 192.168.1.1. It is the most commonly used IP address format but is being gradually replaced by IPv6.
- **IPv6:** A 128-bit address format designed to replace IPv4 due to the exhaustion of IPv4 addresses, e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334.
- **Public and Private IPs:** Public IP addresses are accessible over the internet, while private IP addresses are used within a LAN and are not directly accessible from the internet.
- **Subnetting:** The practice of dividing a network into smaller sub-networks (subnets) to improve management and performance.

Network Devices

- **Router:** Directs data packets between different networks, often connecting a LAN to a WAN like the internet.
- **Gateway:** Gateways can translate one network protocol into another, allowing devices on different types of networks to communicate with each other.
- **Switch:** Connects devices within a LAN, directing data only to the device for which it is intended.
- **Hub:** A basic networking device that broadcasts data to all devices in a network.
- **Firewall:** Monitors and controls incoming and outgoing network traffic based on predetermined security rules.

Network Devices

Modem

NIC

Repeater

Hub

Switch

Router

Bridge

Gateway

Types of Network Devices

Difference Between Bandwidth and Latency

- **Bandwidth:** The maximum rate of data transfer across a network path, typically measured in Mbps or Gbps.

- **Latency:** The time it takes for a data packet to travel from source to destination, usually measured in milliseconds (ms).

Wireless Networking

- **Wi-Fi:** A wireless networking technology that uses radio waves to provide network access.
- **Bluetooth:** A wireless technology standard for exchanging data over short distances.
- **Access Point:** A device that allows wireless devices to connect to a wired network using Wi-Fi or Bluetooth.

NAT (Network Address Translation)

- A technique used by routers to map multiple private IP addresses to a single public IP address and vice versa, allowing devices on a private network to access resources on the internet.